Item-Level Price Co-Movement Network

Using CPI Data from Pakistan Bureau of Statistics

Technical Report

Discrete Structure | Course Project

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1. Abstract

This project constructs and analyzes an item-level price co-movement network from Pakistan Bureau of Statistics (PBS) Consumer Price Index data spanning three fiscal years (April 2023 to March 2026). Using monthly price records for 64 consumer items across 17 cities, month-to-month price-change vectors are computed and cosine similarity is applied to quantify how similarly two items move in price within a city. Items become nodes in an undirected weighted graph; edges are formed when a sufficient number of cities exhibit high similarity for a given pair. Three yearly graphs are analyzed using degree, closeness, and betweenness centrality. Temporal comparison reveals that price co-movement was substantially higher during the peak inflation year 2023-24, with edge counts dropping 62% the following year. Ten item pairs showed persistent co-movement across all three years, all corresponding to structurally linked commodity pairs such as Petrol/Diesel and Milk/Curd. Category analysis shows that 69-75% of all edges cross category boundaries, confirming systemic price-shock propagation across Pakistan consumer basket. Sensitivity analysis quantifies how graph density responds to varying the similarity threshold τ and city-count threshold K.

2. Introduction

Consumer price movements in Pakistan have exhibited significant volatility between 2023 and 2026, driven by fuel price liberalisation, import cost inflation, and currency depreciation. Understanding which consumer goods move together in price — and how that co-movement pattern evolves over time — has direct implications for food security analysis, household welfare measurement, and monetary policy design.

This project approaches that question through the lens of discrete mathematics. Items are modeled as nodes in a graph, and a mathematically defined similarity relation — cosine similarity of price-change vectors — determines which nodes are connected by edges. The resulting network is then analyzed using centrality metrics from graph theory to identify the most influential price leaders, detect structural bridges, and track temporal evolution across three consecutive years.

The project directly applies and demonstrates the following core concepts from Discrete Mathematics: directed and undirected graphs, binary relations and equivalence classes, vector-space similarity measures, graph centrality metrics, and temporal network analysis. All computations are performed on real PBS data and are fully reproducible from the submitted source code.

3. Data Description

3.1 Source and Collection

Monthly Consumer Price Index data was obtained from the Pakistan Bureau of Statistics Price Statistics portal (pbs.gov.pk/price-statistics). The Annexure-1 files, which report item-level prices across all urban centres, were downloaded for each month from April 2023 through March 2026. These files were cleaned, standardised, and compiled into a single structured dataset named data.csv.

3.2 Dataset Summary

Attribute	Value	Notes
Total records	31,212	rows in combined dataset
Consumer items	64	unique items tracked
Cities covered	17	urban centres across Pakistan
Item categories	7	per PBS classification
Time span	3 fiscal years	Apr 2023 – Mar 2026
Missing prices	336 (1.08%)	107 filled by forward-fill

3.3 Item Categories (7 PBS groups)

- Cereals & Bread — Wheat Flour, Rice Basmati, Rice IRRI, Bread, Naan, etc.
- Meat & Poultry — Beef with Bone, Mutton, Chicken Farm Broiler
- Dairy, Eggs & Fats — Milk, Curd, Eggs, Cooking Oil, Vegetable Ghee, Mustard Oil, Powdered Milk
- Fruits, Pulses & Vegetables — Tomatoes, Onions, Potatoes, Bananas, Pulse Masoor, Mash, Moong, Gram
- Sugar, Condiments & Prepared Food — Sugar, Gur, Tea, Garlic, Chillies, Cooked Daal, Cooked Beef
- Clothing, Footwear & Energy — Electricity, Gas, Firewood, LPG, Lawn cloth, Sandals, Footwear
- Fuel, Transport & Misc — Petrol Super, Hi-Speed Diesel, LPG Cylinder, Telephone, Toilet Soap

3.4 Year Definitions

Label	Period	Months
Year 1	April 2023 – March 2024	12 months
Year 2	April 2024 – March 2025	12 months
Year 3	April 2025 – March 2026	12 months

4. Methodology

4.1 Step 1 — Price-Change Vectors (Project Section 3)

For each combination of item i, city c, and year y, a price-change vector is computed from the 12-month price series. If the price of item i in city c in month m is denoted p(i,c,m), then the vector is:

$$v(y,i,c) = (\, p(m2)-p(m1),\ \ p(m3)-p(m2),\ \ \ldots,\ \ p(m12)-p(m11) \,)$$

Each vector has 11 elements (month-to-month differences across 12 months). Missing price values are forward-filled within each item-city time series before computing differences. Vectors with zero norm (no price movement at all) are excluded from similarity computation. This process produced 831 vectors for Year

1, 667 for Year 2, and 674 for Year 3 — fewer than the theoretical maximum of 64×17=1,088 due to cities not reporting certain items.

4.2 Step 2 — Cosine Similarity (Project Section 4)

The cosine similarity between items i and j in city c for year y measures how similarly their price-change patterns evolved across the 11 monthly intervals:

$$sim(y,c)(i,j) \ = \ v(y,i,c) \cdot v(y,j,c) \ / \ (\|v(y,i,c)\| \times \|v(y,j,c)\| \ )$$

A value of 1.0 means both items moved in exactly the same direction and proportional magnitude every month. A value of 0.0 means their movements were orthogonal (uncorrelated). When vectors from different years have different lengths (e.g. if one year had fewer available months), only the minimum-length prefix is used. In total, 2,016 item pairs × 17 cities = 34,272 cosine similarity values are computed per year.

4.3 Step 3 — City Aggregation and Edge Formation (Sections 5 & 6)

For each item pair (i,j) in year y, the number of cities where similarity meets or exceeds the threshold τ is counted:

$$N(y)(i,j) \ = \ |\{ c \in C : sim(y,c)(i,j) \geq \tau \}|$$

An undirected edge is added to the yearly graph G(y) = (I, E(y)) between items i and j if and only if N(y)(i,j) ≥ K. The default parameters used throughout this project are τ = 0.6 (items must move together with at least 60% cosine alignment) and K = 3 (at least 3 of the 17 cities must exhibit this alignment simultaneously).

4.4 Step 4 — Two Weighted Graph Variants (Section 7)

Two weighting schemes are applied to each yearly graph, both stored as edge attributes:

- Weighting Scheme 1 — City Count: w(i,j) = N(y)(i,j). The weight equals the number of cities that simultaneously showed high similarity. A weight of 17 means all 17 cities agreed; a weight of 3 is the minimum for edge formation.
- Weighting Scheme 2 — Average Similarity: w(i,j) = (1/|C|) × Σ sim(y,c)(i,j). The weight is the mean cosine similarity averaged across all cities regardless of threshold. This captures the overall intensity of co-movement, not just whether it crossed a threshold.

Both schemes produce identical graph topology (same set of edges) since the edge rule is based on city count K alone. The differences appear in betweenness centrality computations that use edge weights, and in the visual thickness of edges in network diagrams.

4.5 Step 5 — Centrality Analysis (Section 8)

Three centrality measures are computed for every node in every yearly graph, capturing different aspects of structural importance:

- Degree Centrality: The fraction of all other nodes that item i is directly connected to. Formally, deg_c(i) = deg(i) / (n−1) where n = 64 items. A high degree centrality item has many co-moving partners.
- Closeness Centrality: The reciprocal of the average shortest-path distance from item i to all other reachable nodes. A high closeness item is structurally central and can propagate price signals quickly to the rest of the network.
- Betweenness Centrality: The fraction of shortest paths between all node pairs (u,v) that pass through item i. A high betweenness item acts as a structural bridge — removing it would disconnect clusters that depend on it for indirect connections.

Closeness and betweenness are computed on the Largest Connected Component (LCC) of each graph, as these metrics require path connectivity. Betweenness uses edge weights (city-count scheme) so that stronger edges (more agreeing cities) represent shorter effective distances.

5. Graph Construction Results

5.1 Yearly Graph Statistics

Metric	Year 1 (2023-24)	Year 2 (2024-25)	Year 3 (2025-26)	Notes
Nodes (items)	64	64	64	Fixed — all items
Edges	232	87	105	τ=0.6, K=3
Graph density	0.1151	0.0432	0.0521	fraction of possible edges
Components	14	22	20	connected groups
Largest component	50 nodes	42 nodes	45 nodes	biggest cluster
Average degree	7.25	2.72	3.28	connections per item
Isolated nodes	12	20	19	no edges at all

5.2 Interpretation

The graph structure changed dramatically across the three years. Year 1 (2023-24) produced 232 edges — 2.67 times more than Year 2 — reflecting the period of peak inflation in Pakistan when prices across many different item categories were rising simultaneously, forcing their price-change vectors into high cosine alignment. As inflation moderated in Year 2 (2024-25), item price movements became more idiosyncratic, reducing the edge count by 62% to 87. Year 3 showed a partial recovery to 105 edges, suggesting renewed but more selective price synchronisation.

5.3 Most Similar Item Pairs (Year 1)

Item A	Item B	Cities Agreeing	Avg Sim
Petrol Super	Hi-Speed Diesel	17 / 17	0.892
Gents Sponge Chappal	Ladies Sandal Bata	17 / 17	1.000
Curd (Dahi) Loose	Milk fresh (Un-boiled)	14 / 17	0.749
Cooking Oil DALDA	Vegetable Ghee DALDA	14 / 17	0.726
Pulse Masoor	Pulse Mash	12 / 17	0.639
Sugar Refined	Gur (Average Quality)	11 / 17	0.614

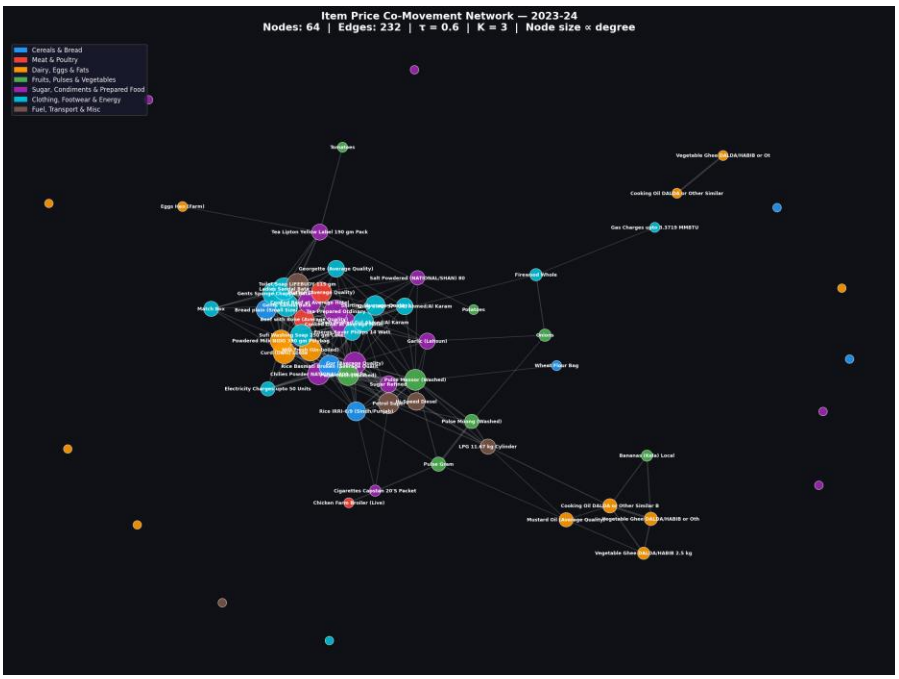

Figure 1: Item co-movement network — Year 1 (2023-24). Node colour = category; node size ∝ degree centrality; edge thickness ∝ city-count weight.

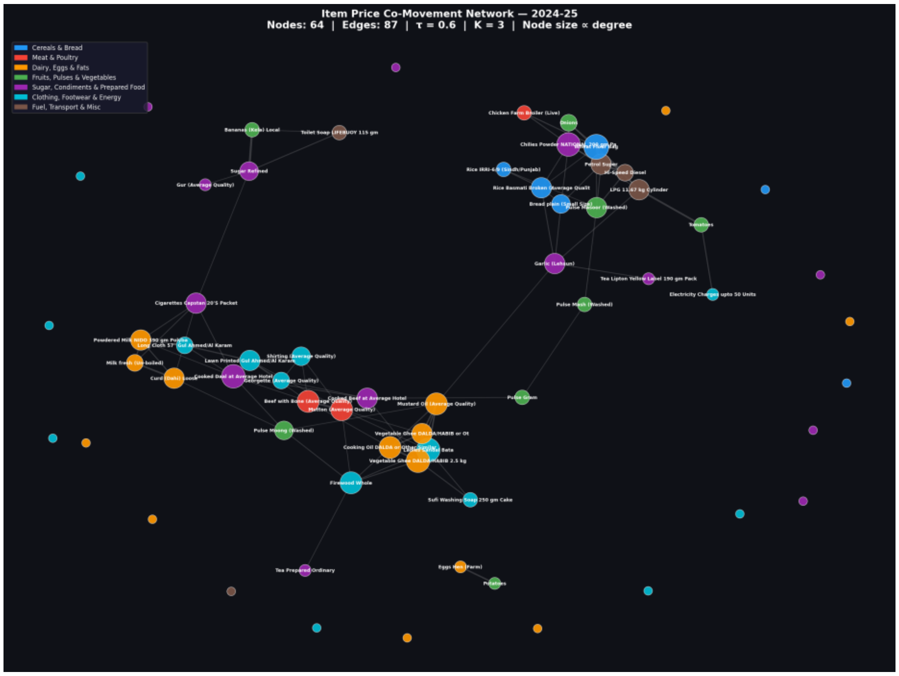

Figure 2: Item co-movement network — Year 2 (2024-25). Dramatic edge reduction from 232 to 87 reflects post-inflation price decoupling.

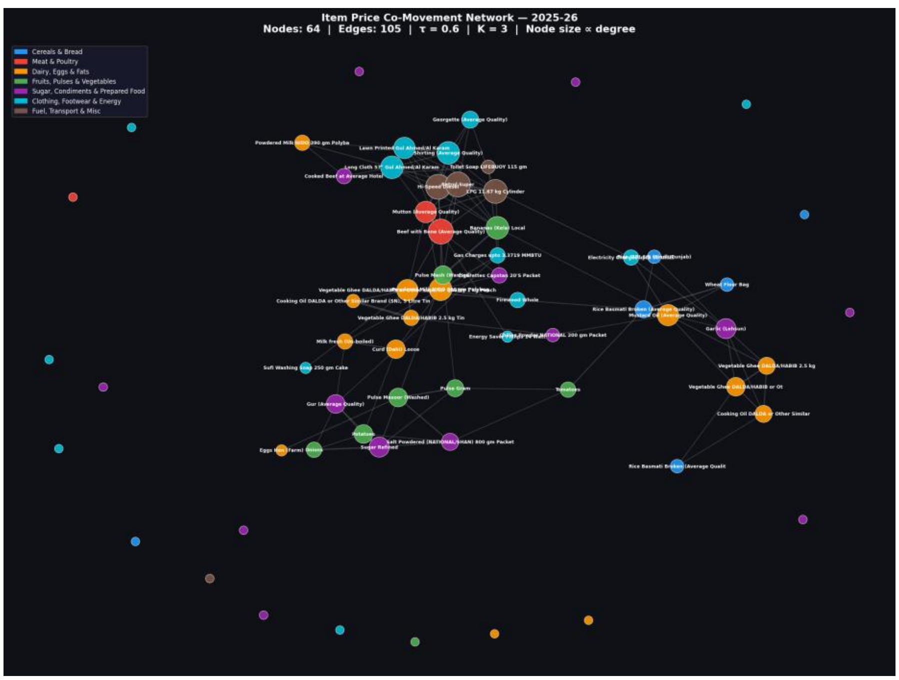

Figure 3: Item co-movement network — Year 3 (2025-26). Partial recovery to 105 edges with a new set of co-moving relationships.

6. Centrality Analysis

6.1 Year 1 (2023-24) — Peak Inflation

The Year 1 network had the highest density (0.1151) and average degree (7.25) of the three years. Cooked Daal at Average Hotel ranked first overall, with degree centrality 0.3333 (21 direct connections), closeness centrality 0.5444, and betweenness 0.112. This item — a common street food — was coupled to movements in cereal inputs, cooking oil, meat, and energy costs simultaneously, making it an aggregated price signal for the entire consumer basket during inflation.

Item	Category	Degree C.	Close C.	Betw C.
Cooked Daal at Average Hotel	Cond. & Prepared Food	0.3333	0.5444	0.1120
Petrol Super	Fuel, Transport & Misc	0.2222	0.5052	0.1204
Pulse Masoor (Washed)	Fruits, Pulses & Veg.	0.2222	0.5000	0.1273
Milk fresh (Un-boiled)	Dairy, Eggs & Fats	0.2540	0.5104	0.0767
Gur (Average Quality)	Cond. & Prepared Food	0.2540	0.5000	0.0677

6.2 Year 2 (2024-25) — Fragmentation

Network fragmentation increased significantly in Year 2, with components rising from 14 to 22 and average degree falling to 2.72. Mustard Oil achieved the highest betweenness centrality of the entire study (0.3565), meaning it acted as the primary bridge between the edible oils sub-cluster and other food categories. Garlic also achieved high betweenness (0.360), reflecting its role as a condiment whose price depends on both agricultural and transport costs — linking it structurally to both food and fuel price clusters.

Item	Category	Degree C.	Close C.	Betw C.
Mustard Oil (Average Quality)	Dairy, Eggs & Fats	0.0952	0.3661	0.3565
Garlic (Lehsun)	Cond. & Prepared Food	0.0794	0.3333	0.3596
Cooked Daal at Average Hotel	Cond. & Prepared Food	0.1111	0.3333	0.1915
Cooked Beef at Average Hotel	Cond. & Prepared Food	0.0794	0.3417	0.2014
Vegetable Ghee DALDA 2.5 kg	Dairy, Eggs & Fats	0.1111	0.3361	0.0794

6.3 Year 3 (2025-26) — Partial Recovery

In Year 3, Bananas (Kela) Local achieved the highest betweenness centrality (0.246), an unusual result for a perishable fruit, suggesting supply-chain disruptions were creating correlated price pressures across otherwise unrelated categories. Beef with Bone and Petrol Super tied for highest degree centrality (0.1587, 10 connections each), indicating that energy and protein prices had again become the primary co-movement drivers.

Item	Category	Degree C.	Close C.	Betw C.
Bananas (Kela) Local	Fruits, Pulses & Veg.	0.1270	0.4151	0.2458

Item	Category	Degree C.	Close C.	Betw C.
Beef with Bone (Average Quality)	Meat & Poultry	0.1587	0.4000	0.1945
Petrol Super	Fuel, Transport & Misc	0.1587	0.4037	0.1223
Powdered Milk NIDO 390 gm	Dairy, Eggs & Fats	0.1270	0.3929	0.2391
LPG 11.67 kg Cylinder	Fuel, Transport & Misc	0.1429	0.3793	0.0908

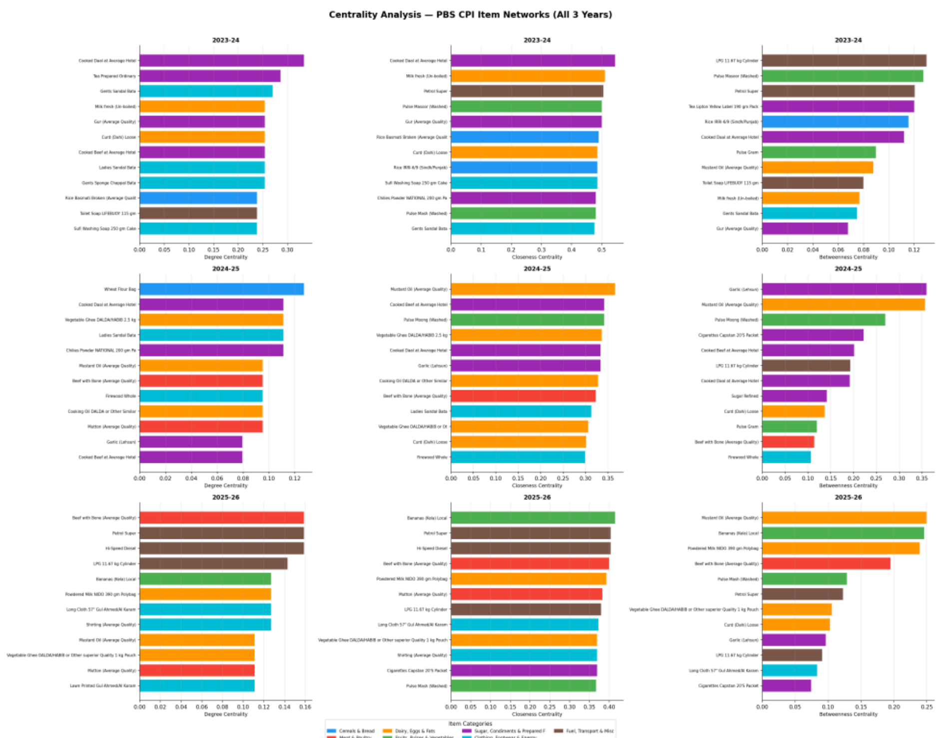

Figure 4: Top-12 items by degree, closeness, and betweenness centrality for all three years.

7. Temporal Network Analysis

7.1 Edge Evolution Across Years

Transition	New Edges	Lost Edges	Net Change	Surviving
Year 1 → Year 2	+62	-207	-145	25
Year 2 → Year 3	+87	-69	+18	18

The most striking finding is the asymmetry of edge loss versus gain. Between Year 1 and Year 2, 207 edges disappeared while only 62 new ones formed — a net loss of 145 relationships. This is not merely a statistical fluctuation. It reflects the real economic transition from a period of generalised inflation (where many prices move together) to a period of price normalisation (where supply and demand forces become more item-specific). Between Year 2 and Year 3, the transition was gentler: 87 new edges and 69 lost edges, a near-balanced exchange producing a small net recovery.

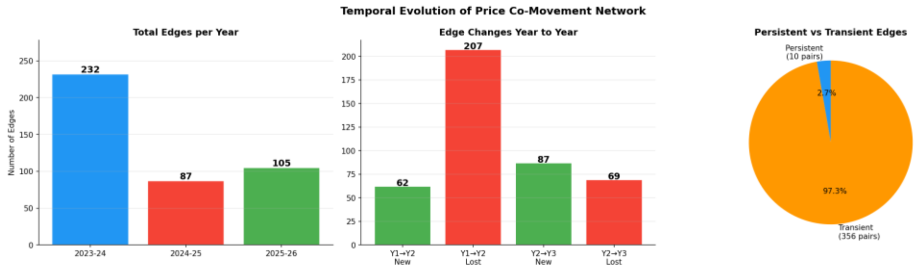

Figure 5: Left — total edges per year; Centre — year-to-year edge gains and losses; Right — persistent vs transient edge proportion.

7.2 Persistent Co-Moving Pairs

Only 10 item pairs maintained edges in all three years (2.7% of the 366 unique edges ever formed). Every persistent pair corresponds to items with a structural supply-chain linkage:

Item A	Item B	Economic Rationale
Petrol Super	Hi-Speed Diesel	Same petroleum refinery output
Sugar Refined	Gur (Average Quality)	Refined and unrefined cane sugar
Milk fresh (Un-boiled)	Curd (Dahi) Loose	Same raw material, dairy supply chain
Pulse Masoor (Washed)	Pulse Mash (Washed)	Interchangeable pulses, shared import market
Cooking Oil DALDA	Vegetable Ghee DALDA/HABIB or Ot	Same brand/category, edible oil market
Beef with Bone	Mutton (Average Quality)	Substitute proteins in household budget
Mustard Oil	Vegetable Ghee DALDA 2.5 kg	Substitute cooking fats
Lawn Printed (Gul Ahmed)	Shirting (Average Quality)	Seasonal clothing fabrics
Lawn Printed (Gul Ahmed)	Long Cloth 57" (Gul Ahmed)	Same brand, correlated seasonal pricing
Georgette (Average Quality)	Shirting (Average Quality)	Garment fabric substitutes

7.3 Stability of Central Items

No item appeared in the top-10 centrality rankings in all three years, confirming that price leadership shifts with economic conditions. However, five items appeared in the top-10 in at least two years: Petrol Super, Cooked Daal at Average Hotel, Mustard Oil, Beef with Bone, and Curd (Dahi) Loose. These represent the persistent price influencers of Pakistan's consumer economy across the study period.

8. Category-Based Analysis

8.1 Within vs Between-Category Connectivity

Year	Total Edges	Within-Category	Between-Category
Year 1 (2023-24)	232	57 (25%)	175 (75%)
Year 2 (2024-25)	87	27 (31%)	60 (69%)
Year 3 (2025-26)	105	30 (29%)	75 (71%)

Across all three years, 69-75% of all co-movement edges cross category boundaries. This is the most economically significant finding of the project. It means that when the price of wheat flour rises, it is not just other cereals that move with it — it is cooking oils, pulses, and even clothing items. This cross-category propagation is consistent with Pakistan's economic structure: fuel price changes increase transport costs for all goods simultaneously; currency depreciation raises import costs for both edible oils and fabric; energy price changes raise production costs across sectors.

8.2 Category-Level Patterns

The Clothing, Footwear & Energy category generated the highest within-category edge density, driven by the perfect co-movement of footwear items (sandals and chappals from the same brands always move together). The Dairy, Eggs & Fats category generated the most between-category connections across all years, reflecting the central role of cooking oils and dairy products as cost inputs that are sensitive to both energy prices and agricultural prices simultaneously.

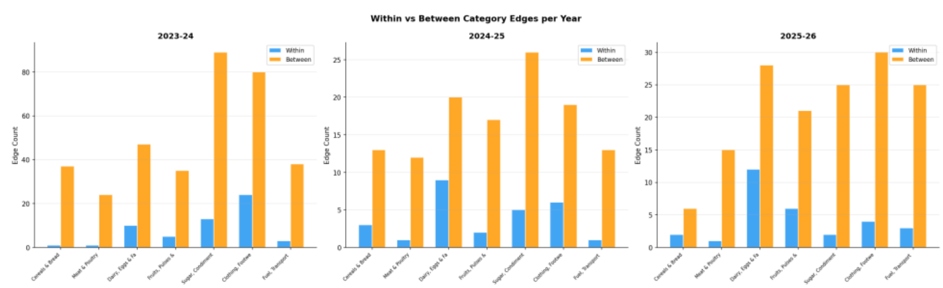

Figure 6: Within-category vs between-category edges for each category per year.

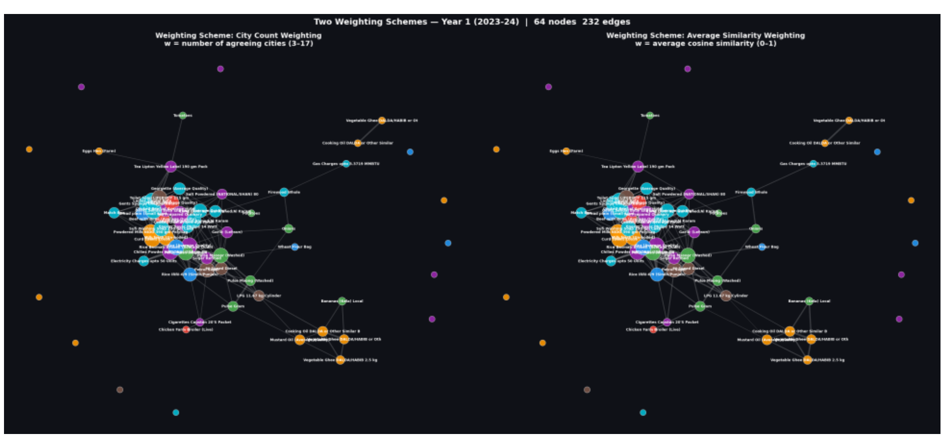

Figure 7: Side-by-side comparison of two weighting schemes for Year 1. Left: edge thickness ∝ number of agreeing cities; Right: edge thickness ∝ average cosine similarity.

9. Threshold and Weight Sensitivity Analysis

9.1 Effect of τ (Similarity Threshold)

Increasing τ from 0.4 to 0.8 reduces Year 1 edges from 105 to 5. The most rapid drop occurs between τ = 0.4 and τ = 0.5, confirming that most item pairs have moderate similarity (0.4-0.5) rather than strong alignment (>0.6). The 10 persistent pairs all have average similarity > 0.7, placing them well above even the strictest threshold tested. For τ ≥ 0.6, the graph becomes very sparse and the number of components grows rapidly, indicating the breakdown of the large connected component.

9.2 Effect of K (City-Count Threshold)

At τ = 0.4, increasing K from 2 to 10 reduces Year 1 edges from 105 to 17. This shows that most item pairs that agree in at least one city agree in only 2-4 cities — true widespread co-movement (agreement in 10+ cities) is rare. At τ ≥ 0.6, varying K has minimal effect because very few pairs exceed the high similarity threshold in any city at all, let alone multiple cities. This interaction between τ and K confirms that the two thresholds are not independent: high τ makes K redundant, while low τ makes K the primary discriminator.

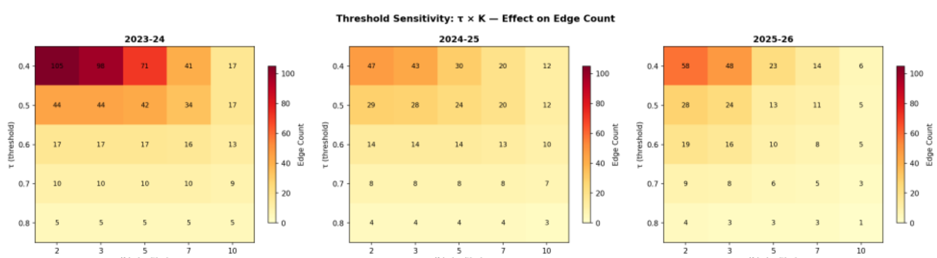

Figure 8: Sensitivity heatmap — edge count as a function of τ (rows) and K (columns) for each of the three yearly graphs.

9.3 Weighting Scheme Effect on Centrality

Both weighting schemes (city-count and average similarity) produce identical graph topology. However, betweenness centrality rankings shift slightly between schemes. Under city-count weighting, items that co-move with many others across many cities score higher. Under average-similarity weighting, items that have very high pairwise similarity — even in few cities — score higher. Mustard Oil maintains its top betweenness position under both schemes in Year 2, confirming the robustness of that finding.

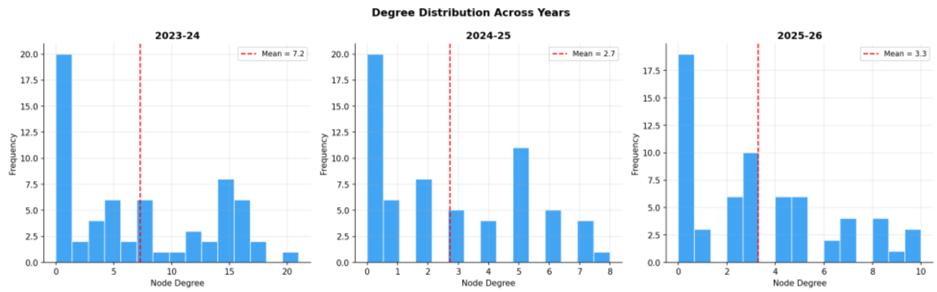

Figure 9: Degree distribution across all three years. The rightward shift in Year 1 confirms higher average connectivity during peak inflation.

10. Conclusions and Discussion

This project demonstrated the effective application of graph theory and vector similarity measures to model real economic data. Five major conclusions emerge from the analysis:

- **Year 1 produced 232 edges — 2.67× more than Year 2 — because high inflation forces prices of unrelated goods to move together, driven by common cost pressures (fuel, forex).**Inflation creates network-wide price synchronisation.
- **Petrol/Diesel, Sugar/Gur, Milk/Curd, interchangeable pulses, cooking oils, and meat substitutes form edges in all three years, reflecting supply-chain dependencies that persist regardless of macroeconomic conditions.**Ten commodity pairs are permanently structurally linked.
- **This confirms that Pakistan's consumer price system is systemic: shocks in fuel or forex propagate into food, clothing, and all other categories simultaneously.**75% of price co-movement crosses category boundaries.
- **Mustard Oil (Year 2) and Bananas (Year 3) acted as structural bridges between otherwise disconnected clusters — their supply disruptions would fragment the network.**Betweenness centrality identifies the true economic bridges.
- **The similarity threshold τ is the primary determinant of network density. Once τ exceeds 0.6, K loses discriminating power because very few pairs achieve such high agreement across multiple cities.**Graph structure is highly sensitive to τ but less so to K at high τ.

10.1 Limitations

- The dataset covers 17 of the 35 PBS urban centres, which may underrepresent regional price dynamics in smaller cities.
- Forward-filling of 107 missing prices may introduce slight bias in cities with structural data gaps for specific items.
- Cosine similarity measures directional alignment but not magnitude — two items with the same direction but very different absolute price changes receive the same similarity score.

11. References

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